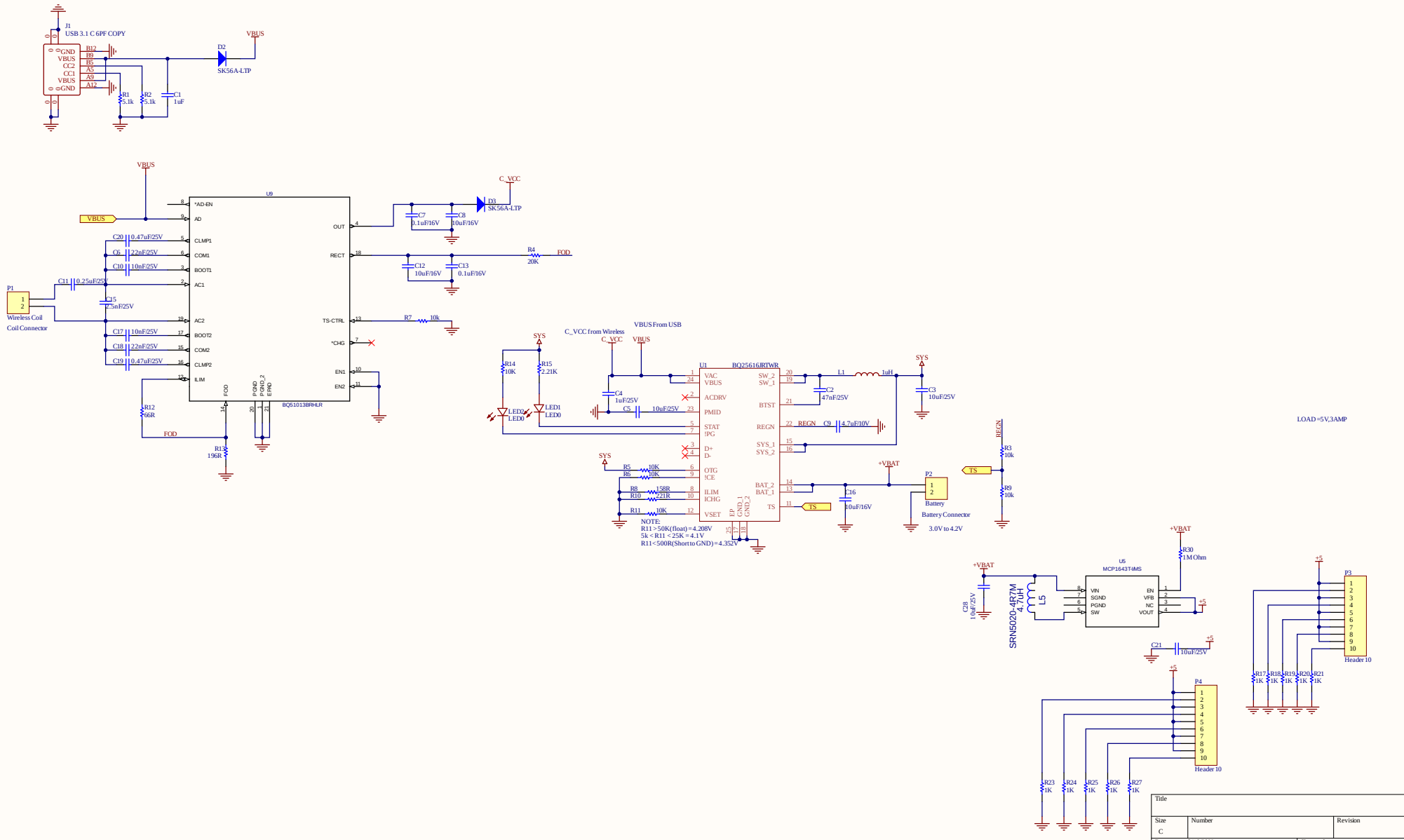
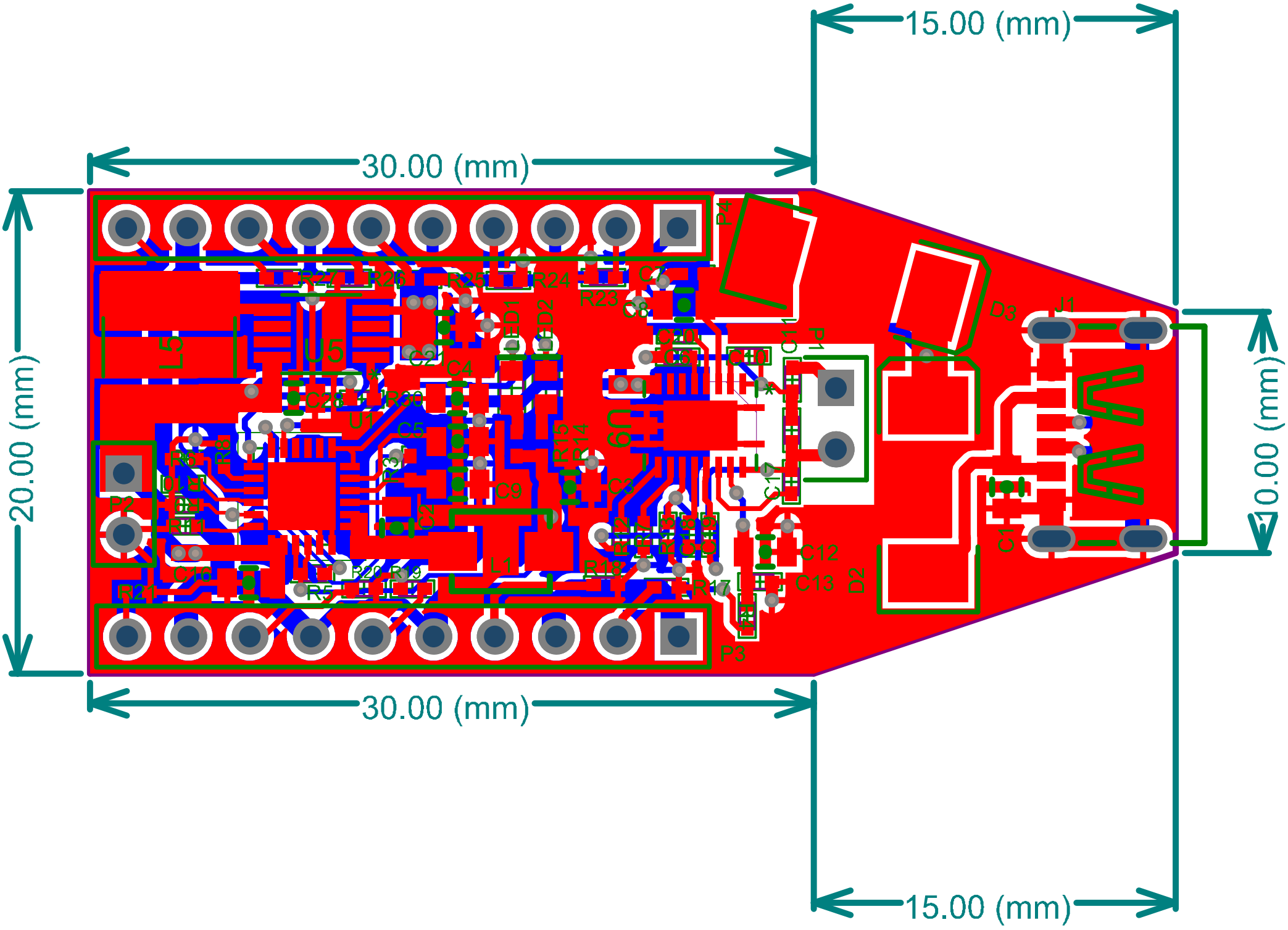

J1
USB 3.1 C 6PF COPY
GND
VBUS
CC2
CC1
VBUS
GND
B12
B9
B5
A5
A9
A12
R1
5.1k
R2
5.1k
C1
1uF
D2
SK56A-LTP
VBUS
U9
*AD-EN
AD
CLMP1
COM1
BOOT1
AC1
AC2
BOOT2
COM2
CLMP2
ILIM
FOD
PGND
PGND_2
EPAD
BQ51013BRHLR
OUT
RECT
TS-CTRL
*CHG
EN1
EN2
C_VCC
D3
SK56A-LTP
C7
0.1uF/16V
C8
10uF/16V
C12
10uF/16V
C13
0.1uF/16V
R4
20K
FOD
R7
10k
C20
0.47uF/25V
C6
22nF/25V
C10
10nF/25V
C11
0.25uF/25V
C15
2.5nF/25V
C17
10nF/25V
C18
22nF/25V
C19
0.47uF/25V
R12
66R
R13
196R
P1
Wireless Coil
Coil Connector
SYS
R14
10K
R15
2.21K
LED2
LED0
LED1
LED0
C_VCC from Wireless
VBUS From USB
C_VCC
VBUS
U1
BQ25616JRTWR
VAC
VBUS
ACDRV
PMID
STAT
!PG
D+
D-
OTG
!CE
ILIM
ICHG
VSET
EP
GND_1
GND_2
SW_2
SW_1
BTST
REGN
SYS_1
SYS_2
BAT_2
BAT_1
TS
L1
1uH
C2
47nF/25V
C3
10uF/25V
C9
4.7uF/10V
C4
1uF/25V
C5
10uF/25V
C16
10uF/16V
R5
10K
R6
10K
R8
158R
R10
221R
R11
10K
NOTE:
R11>50K(float)=4.208V
5k<R11<25K=4.1V
R11<500R(Shortto GND)=4.352V
+VBAT
P2
Battery
Battery Connector
3.0V to 4.2V
REGN
R3
10k
R9
10k
LOAD=5V,3AMP
+VBAT
R30
1M Ohm
U5
MCP1643T4MS
VIN
SGND
PGND
SW
EN
VFB
NC
VOUT
C28
10uF/25V
SRN5020-4R7M
4.7UH
L5
C21
10uF/25V
P3
Header 10
P4
Header 10
R17
1K
R18
1K
R19
1K
R20
1K
R21
1K
R23
1K
R24
1K
R25
1K
R26
1K
R27
1K
Title
Size
C
Number
Revision
Date:
3/19/2022
Sheet of
File:
C:\Users\..\BQ51013BYFPR_Circuit.SchDoc
Drawn By:


15.00 (mm)
30.00 (mm)
20.00 (mm)
10.00 (mm)
30.00 (mm)
15.00 (mm)
P1
P2
P3
J1
D3
D2
U5
U6
U1
L5
L1
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
C13
C14
C15
C16
C17
C18
C19
C20
C21
C22
C23
C24
C25
C26
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
R15
R16
R17
R18
R19
R20
R21
R22
R23
R24
R25
R26
D1
D2
Q1